

Business
Value

AI system
Thinking

End-to-end

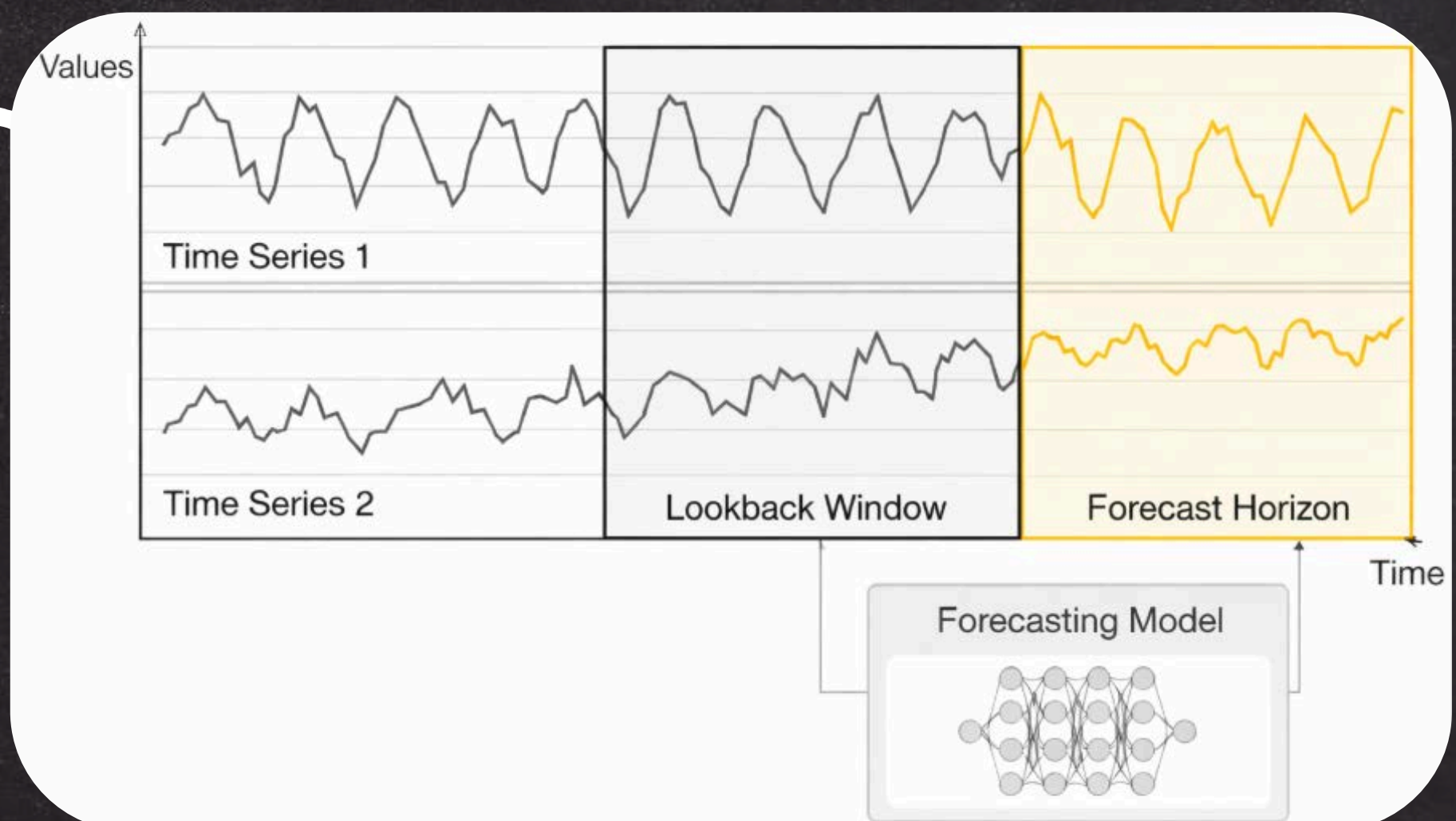
**Explainable AI
System for**

**Time Series Forecasting
Under Uncertainty**

By Mita RAKOTONIAINA



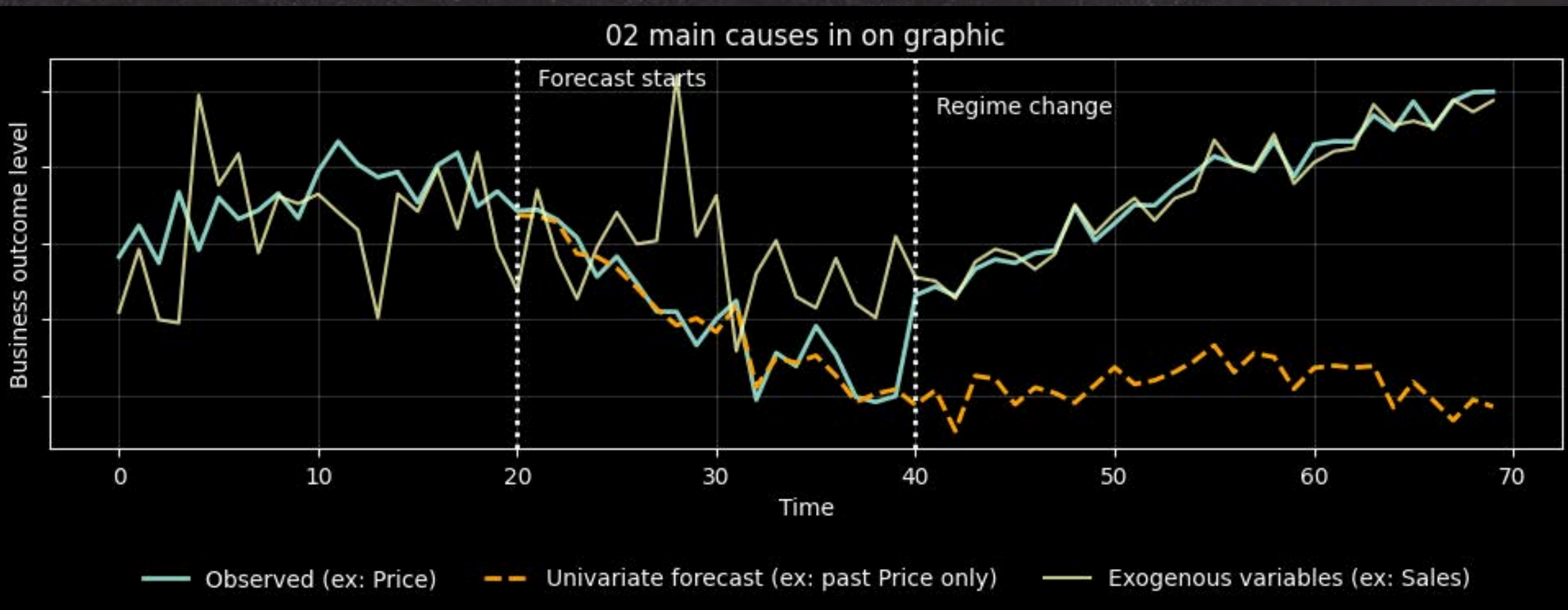
What is Time Series Forecasting ?



We observe the past
We anticipate the future
We decide under uncertainty



1. When does forecasting fail in a system?

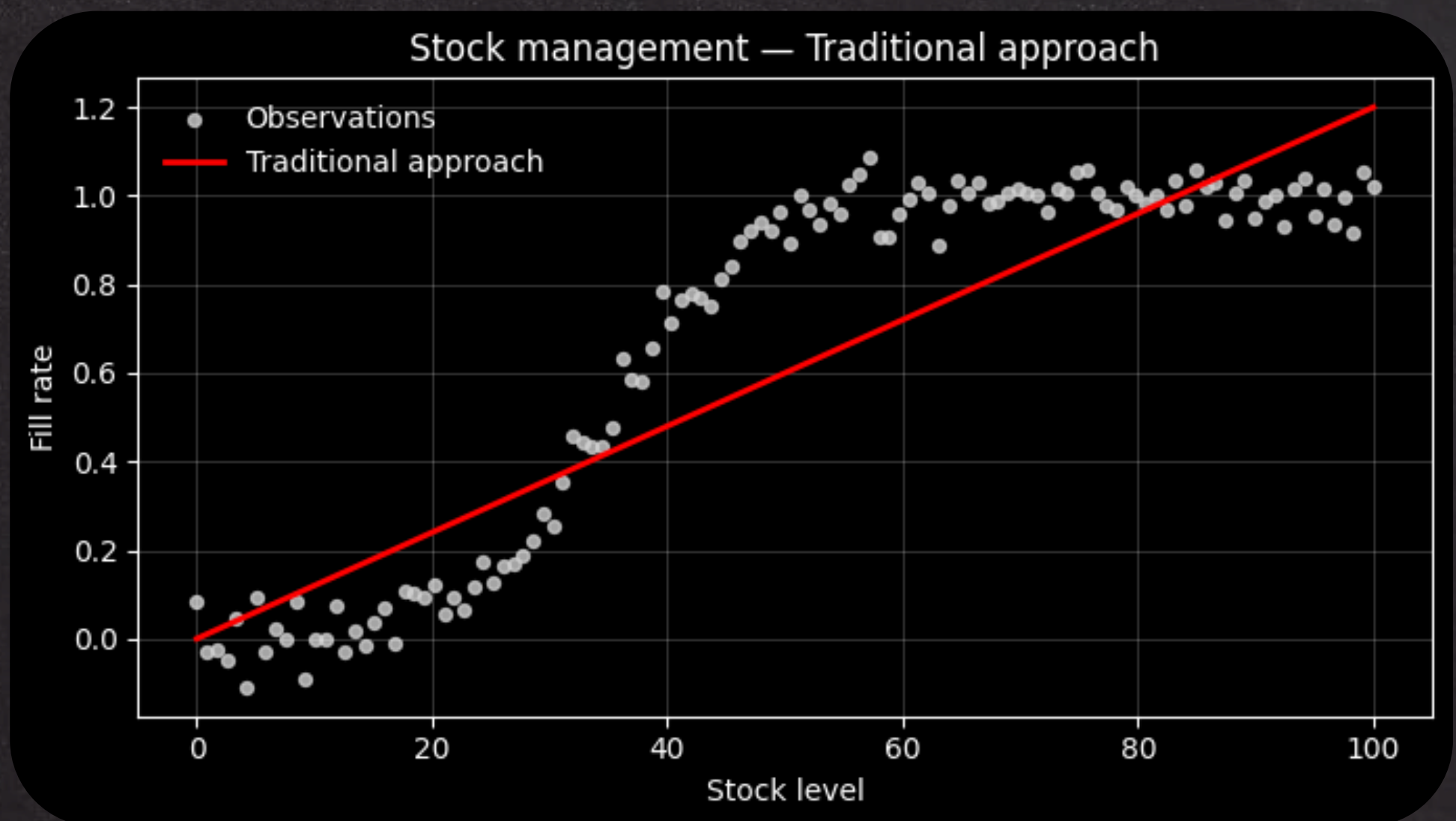


Interactions are hidden

Regime changes occur



1. Why does forecasting fail in real-system?

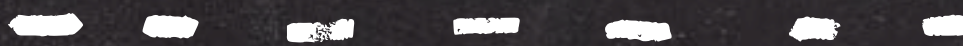


Because traditional models assume linear and stable relationships. Real system doesn't





Why does it matter ?



These two assumptions hide thresholds and tipping points → delayed or excessive decisions

Marketing : Proportional budget allocation → ineffective spend or premature stop

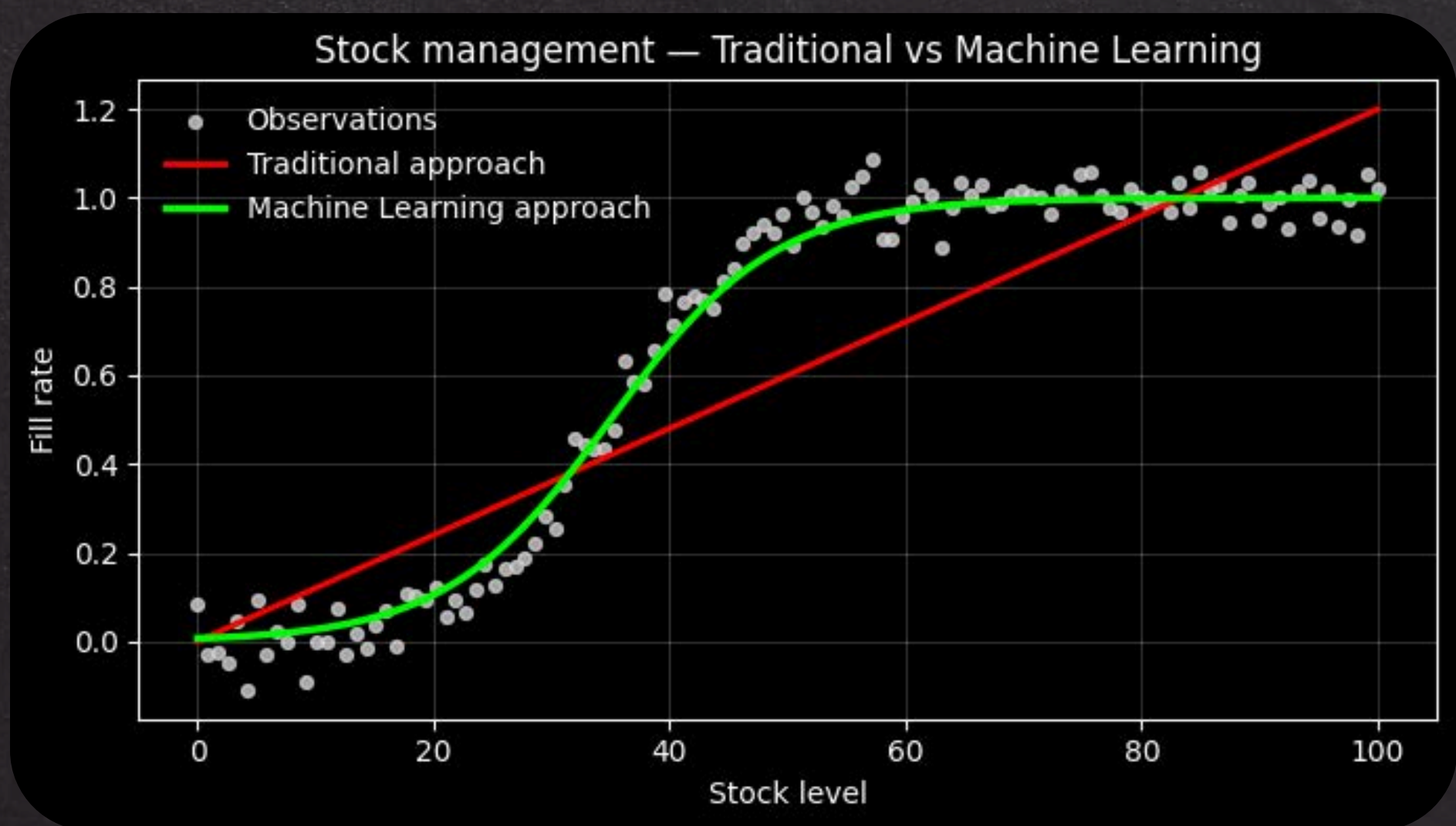
Inventory management : “Average” stock levels → frequent or costly stockouts

Pricing, Human Resources, Corporate Finance, and more...



2.

ML system as the solution

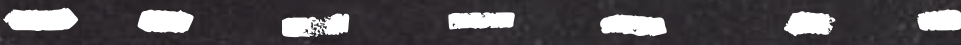


Machine Learns relationship :

- Linear or not
- Threshold-based
- Evolving across regimes



2. ML adoption : what limits ?



A 2025 survey by IBM & Oxford Economics on 2000 organizations worldwird shows trust as a major (46%) barrier to ML adoption.

Because of :

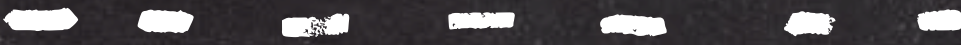
- Opaque decision logic
 - Unclear data reliability
 - Bias concerns
- 

→ Limit of production deployment



2.

Explainable AI (XAI) : restoring trust in ML



XAI enables :

- understanding why a decision was made
- Detecting and correcting errors or biases

As a result :

- Decisions are auditable
 - Results can be questioned or adjusted
 - The stochastic nature of AI systems is controlled, making them more predictable
 - ML can be safely deployed in production
- 



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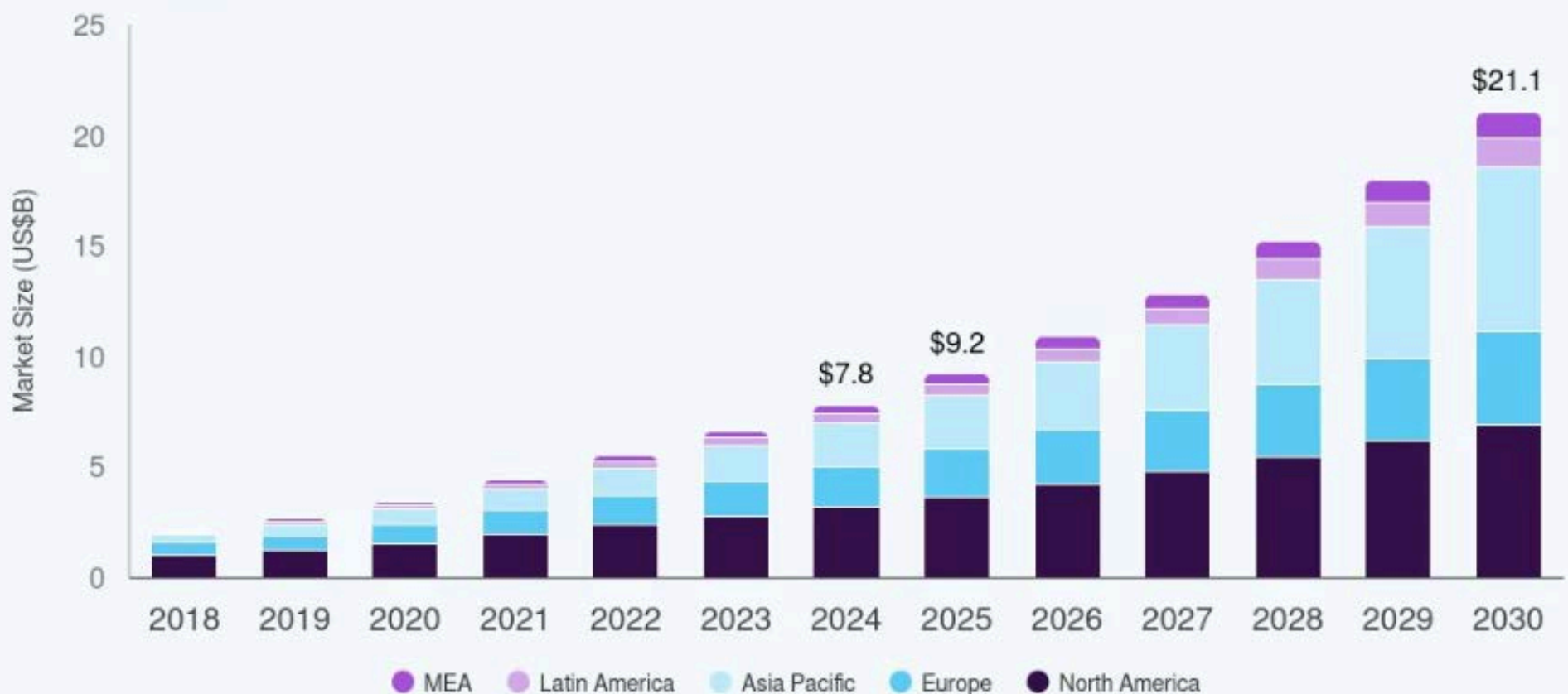
Solution
Value

2.

Explainable AI :

Explainable AI Market

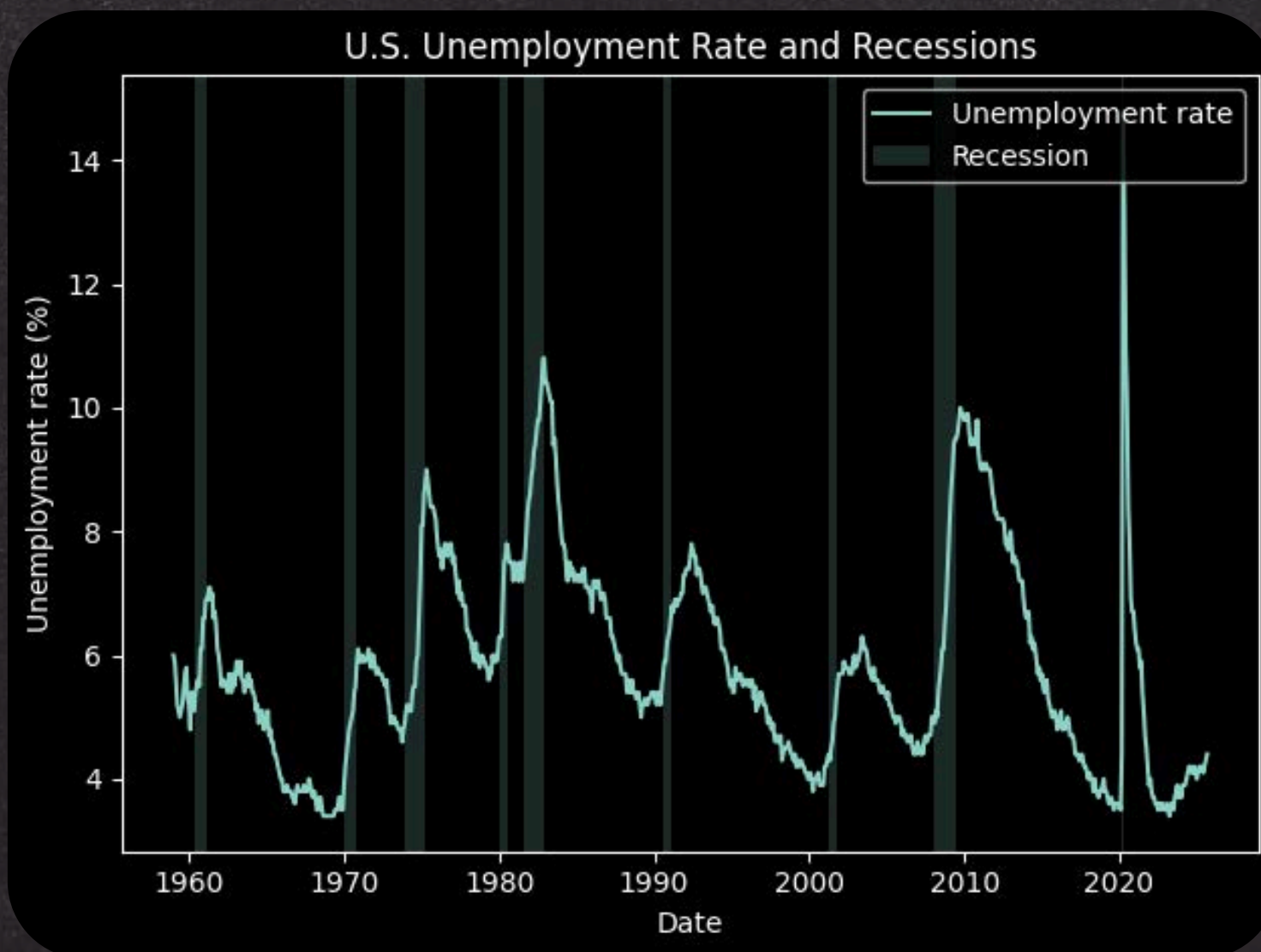
Size, by Region, 2018 - 2030



Explainability is required by regulation (EU AI, BIPA, GAI Measures, ...) and is becoming a market standard.



3. USE Case



- U.S. economy is driven by global shocks (oil crises, 2008 financial crisis, Covid-19) → Multivariate approach.
- Stability and Linearity fail when decisions matter most → AI explainable



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3. Approach

We use two complementaries AI :



MACHINE LEARNING

1

MULTIVARIATE
FORECASTING
MANAGE UNCERTAINTY
EXPLAINABILITY



GENERATIVE AI

2

REPORT RESULTS
INTERACT WITH
DECIDORS / ANALYSTS

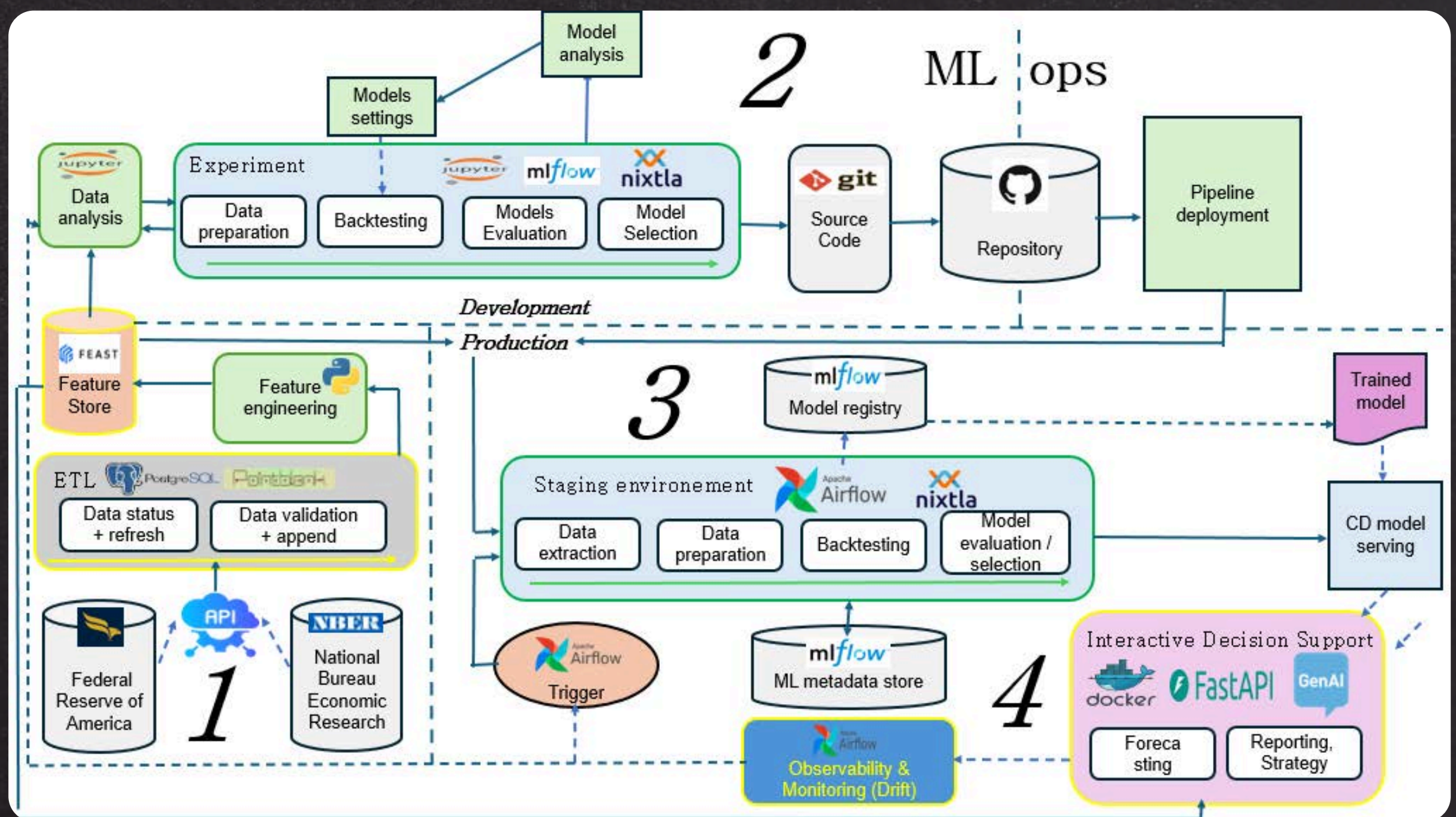


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4.

AI system Workflow



Each part of this workflow — and how they interact — will be explained over the next month.



Outline



DATA PIPELINE

1

PREPARE DATA FOR
EXPLAINABLE AI



PROTOTYPING

2

CREATE AN
EXPLAINABLE AI
SYSTEM THAT
SOLVES THE
BUSINESS PROBLEM."



STAGGING

3

ENSURE A SAFE
TRANSITION TO
PRODUCTION.



PRODUCTION

4

ENSURE THE
DEPLOYMENT,
MONITORING AND
DECISION SUPPORT

